

Digital Forensics & Incident Response

UNIT 1 - Foundations of Digital Forensics

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Digital Forensics & it's types



What is Digital Forensics

Digital Forensics is a branch of forensic science that involves the systematic identification, collection, preservation, examination, analysis and reporting of information stored or transmitted in digital form.

The objective of digital forensics is to determine **what happened, when it happened, how it happened and what evidence supports the findings**. The investigation must be performed using reliable and repeatable methods while protecting the integrity of the original evidence.

NIST describes digital forensics as the application of science to the identification, collection, examination and analysis of data while preserving information integrity and maintaining a strict chain of custody.

In simple words

Digital forensics converts **digital traces into reliable investigative evidence**.

Example:

If an employee is suspected of copying confidential files to a USB drive, digital forensics can help determine whether the USB was connected, which files were accessed, when the activity occurred and what other evidence supports the event.



Need for Digital Forensics

Why is Digital Forensics Important?

Modern activities increasingly depend on computers, smartphones, networks, cloud services and digital applications. As a result, incidents such as cyberattacks, fraud, data theft and unauthorised access often leave digital traces.

Digital forensics provides a structured method for discovering and interpreting these traces. It can help investigators reconstruct events even when an attacker has attempted to delete or hide evidence.

Digital forensics can help answer:

Who performed an activity?

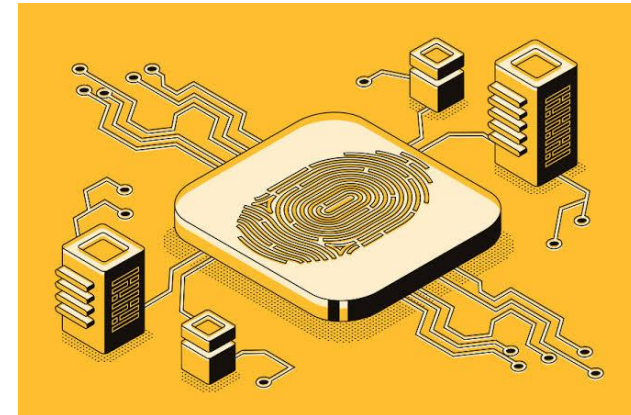
What activity was performed?

When did it happen?

Where did it originate?

How was the activity performed?

What evidence supports the conclusion?



NIST notes that digital evidence can originate from computers, hard drives, mobile devices and many other sources, while investigators must also deal with large volumes of information and the need to avoid altering evidence.



Characteristics of Digital Forensics

Digital forensics is different from ordinary data analysis because the information being examined may ultimately need to support an investigation or legal proceeding.

Scientific

Investigators should use systematic and scientifically defensible methods rather than assumptions.

Evidence-oriented

The objective is to identify information that can establish or disprove relevant facts.

Repeatable

Another competent investigator should be able to understand the procedure and reproduce the result.

Integrity-focused

The original evidence must be protected against unnecessary alteration.

Documented

Actions performed during acquisition, examination and analysis must be properly recorded.

Legally aware

Investigators must work within the applicable authority, procedures and legal framework.



Scope of Digital Forensics

Digital forensics is not limited to analysing a computer's hard disk. Evidence can exist across multiple digital environments.

Major sources include:

Computers and Servers

Files, operating-system artefacts, logs, user activity and application data.

Mobile Devices

Messages, calls, applications, media and other device information.

Networks

Network packets, connections, DNS requests, firewall logs and other network activity.

Cloud Systems

Cloud storage, authentication records, application logs and service activity.

Removable Devices

USB drives, memory cards and external storage.

Email Systems

Headers, attachments, routing information and authentication information.



Major Types of Digital Forensics

1. Disk Forensics

Disk forensics involves examining data stored on physical or logical storage media.

It may include file systems, deleted files, metadata, user activity and operating-system artefacts.

2. Memory Forensics

Memory forensics examines volatile information stored in RAM.

It can provide evidence about running processes, network connections and other information that may disappear when a system is powered off.

3. Network Forensics

Network forensics examines network communications to understand connections, traffic patterns, transferred information and possible malicious activity.

4. Mobile Forensics

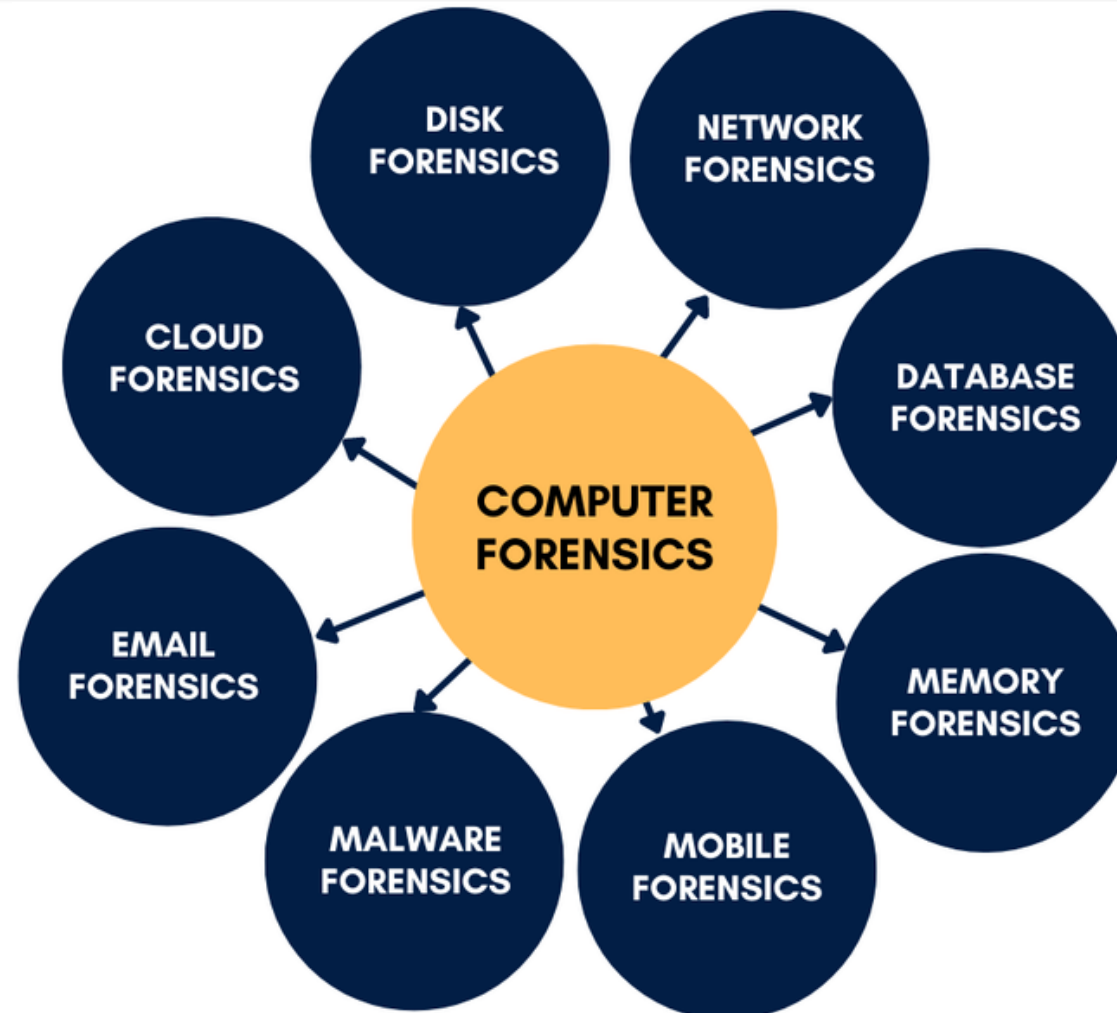
Mobile forensics involves the examination of smartphones and tablets to identify relevant application, communication and device artefacts.

5. Email Forensics

Email forensics examines email content and metadata, particularly headers and routing information, to determine the origin, path and authenticity of messages.



Major Types of Digital Forensics



Digital Evidence

Digital evidence is information stored or transmitted in digital form that has potential value in an investigation. Unlike physical evidence, digital evidence can be copied, modified or deleted very easily. It can also exist in several locations simultaneously.

Examples

Files and folders, Emails, Browser history,
System logs, Application records, Network packets,
Images and videos, Registry artefacts,
Memory contents, Metadata.

Important distinction

Data becomes useful evidence when its relevance, origin, integrity and context can be established.



Digital Evidence: A Simple Example

Consider a suspected data-theft incident.

An investigator discovers:

A confidential document was accessed at 10:30 AM.

This information alone does not establish who copied the document.

Additional evidence may show:

USB device connected at 10:32 AM.

Another artefact may show:

File accessed again at 10:33 AM.

A network record may show:

Connection to another system at 10:35 AM.

The investigator can correlate these pieces of evidence to reconstruct what happened.

Key concept

Digital forensics rarely depends on one artefact.

A strong finding is generally supported by **multiple related sources of evidence.**



Forensic Readiness

Forensic readiness is the ability of an organisation to collect, preserve and use digital evidence effectively when an incident occurs.

Instead of preparing only after an attack has happened, an organisation establishes appropriate technical and procedural capabilities beforehand.

Why is it required?

- During an incident, investigators may need information such as:
- Authentication records
- Endpoint activity
- Network activity
- System logs
- User activity
- File-access information

If these records were never collected or have already been deleted, investigation becomes much more difficult.



Components of Forensic Readiness

Forensic readiness requires both **technical preparation** and **organisational preparation**.

Technical preparation

Systems should generate and retain relevant logs, while important systems should maintain consistent time settings.

Evidence protection

Logs and other evidence should be protected against unauthorised modification or deletion.

Policies and procedures

The organisation should define how evidence is collected, preserved, transferred and accessed.

Trained personnel

Employees responsible for incident response and evidence handling should understand their responsibilities.

Legal coordination

Organisations should know when legal, compliance or law-enforcement involvement is required.

Objective

The purpose is to ensure that when an incident occurs, **useful evidence is already available and can be handled properly**.



What is NIST?

The **National Institute of Standards and Technology (NIST)** is a U.S. government agency that develops **standards, guidelines, frameworks, and technical publications** to improve cybersecurity, information security, privacy, and digital investigation practices.

In cybersecurity and digital forensics, NIST provides structured guidance that helps organisations and investigators follow **consistent, repeatable, and technically sound procedures**.

NIST provides guidance for areas such as:

- **Digital evidence collection and preservation**
- **Forensic examination and analysis**
- **Incident response**
- **Evidence handling and documentation**
- **Forensic tools and techniques**
- **Cybersecurity and risk management**

Important NIST Resources

NIST SP 800-86: *Guide to Integrating Forensic Techniques into Incident Response*

NIST SP 800-61: *Computer Security Incident Handling Guide*

NIST Cybersecurity Framework (CSF): Provides a structured approach to managing cybersecurity risks.



How is NIST Useful for DFIR?

NIST helps DFIR teams establish a **structured and consistent approach** to investigating security incidents and handling digital evidence. Its guidance helps investigators understand what should be collected, how evidence should be handled, how incidents should be investigated, and how findings should be documented.

Key Uses in DFIR

- 1. Incident Investigation:** Provides structured approaches for identifying, analysing and responding to cybersecurity incidents.
- 2. Evidence Collection:** Guides investigators in identifying and acquiring relevant digital evidence while maintaining its integrity.
- 3. Evidence Preservation:** Helps ensure that evidence is properly protected, documented and handled throughout an investigation.
- 4. Forensic Analysis:** Provides methodologies for examining digital artefacts and extracting information relevant to an investigation.
- 5. Documentation & Reporting:** Encourages proper documentation of investigative actions, findings and conclusions.



Locard's Exchange Principle



Locard's Exchange Principle

Locard's Exchange Principle is a fundamental principle of forensic science stating that when two entities come into contact, an exchange or transfer of material can occur between them.

In traditional forensic investigation, a person entering a crime scene may leave something behind and may also take something from the scene.

The principle can be applied conceptually to digital investigations because interaction with a digital system can create traces.

Simple idea

Interaction → Trace

For example:

User logs into system → Login record

USB connected → USB-related artefact

Website visited → Browser artefact

File accessed → File-system/application trace



Locard's Principle in Digital Forensics

Digital systems continuously generate records as users and applications interact with them.

For example, when a user connects a USB device, Windows may create artefacts associated with that device. Similarly, authentication activity can generate login records and browsing activity can generate browser history.

These traces may not always provide a complete picture individually, but they can become valuable when correlated with other evidence.

Examples

USB Connection: May leave device-related artefacts.

Login Activity: May leave authentication records.

Browser Activity: May leave history, cache and download information.

File Activity: May produce metadata and application artefacts.

Important point

Locard's principle helps investigators **look for traces of interaction**; it does not by itself prove who performed an action.



Example: Applying Locard's Principle

Scenario

An employee is suspected of transferring confidential information to a USB drive.

The investigator finds:

23:40 — Confidential document accessed

23:42 — USB device connected

23:44 — File accessed again

23:47 — USB device disconnected

These observations create a possible relationship between the events.

However, the investigator should not immediately conclude that data was copied.

Further examination may include:

USB artefacts, file metadata, operating-system logs and other relevant records.

Conclusion

Locard's principle helps establish where investigators should look for traces, while corroborating evidence establishes what actually happened.



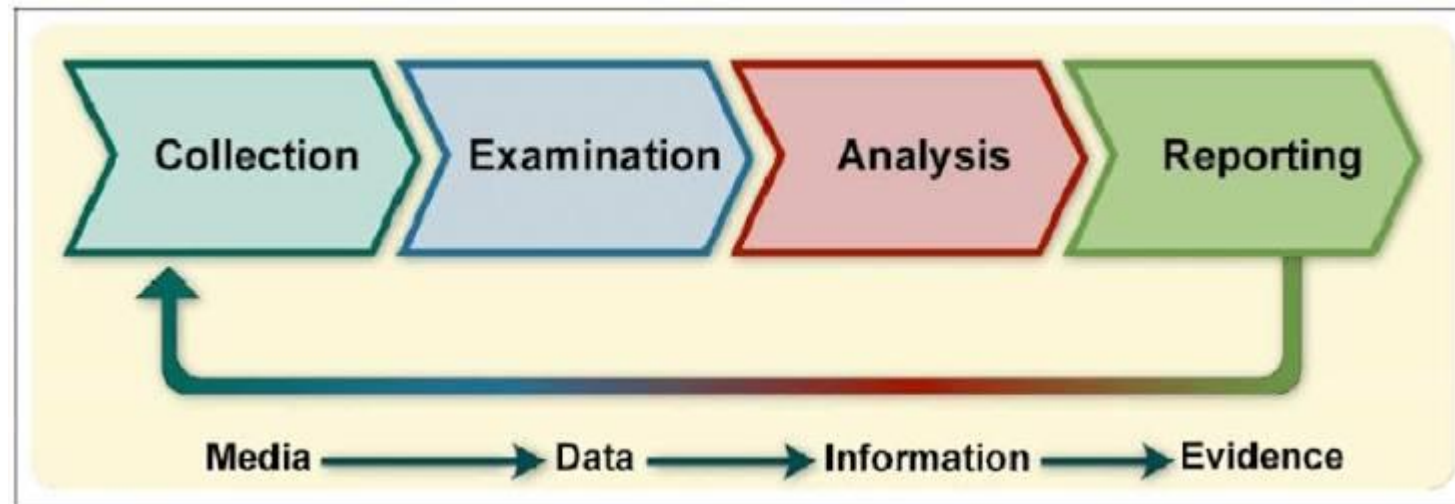
Computer Forensic Investigation Process

A **computer forensic investigation process** is a structured sequence of activities used to identify relevant evidence, acquire and preserve it, examine and analyse the information, and communicate the findings.

The syllabus simplifies this process into four major stages:

Assess → Acquire → Analyse → Report

NIST's forensic guidance similarly describes the major phases as **collection, examination, analysis and reporting**.



Stage 1: Assess the Situation

Assessment is the initial stage in which investigators understand the incident, define its scope and determine what evidence may be relevant.

Before collecting evidence, investigators should understand the circumstances surrounding the incident.

Important questions

- What happened?
- When was it discovered?
- Which systems may be affected?
- Which users or accounts are involved?
- What evidence could answer the investigation questions?
- Is the system currently running?

Why this matters

An investigation without clear scope can result in unnecessary evidence collection, wasted time and potential privacy or legal problems.



Stage 2: Acquire the Data

Acquisition is the process of obtaining relevant data from potential evidence sources while following procedures designed to preserve its integrity.

Examples include:

- Creating forensic images
- Capturing volatile memory
- Collecting relevant logs
- Obtaining network captures
- Acquiring data from removable devices

The acquisition process should be documented so that another investigator can understand what was collected and how it was obtained.

NIST defines collection as identifying, labelling, recording and acquiring relevant data while following procedures that preserve data integrity.



Stage 3: Analyse the Data

Analysis is the process of interpreting examined evidence to determine its significance and answer the questions that led to the investigation.

Analysis does not simply mean opening files and looking at them.
Investigators may need to:

- Correlate different artefacts
- Compare timestamps
- Reconstruct activity
- Identify suspicious behaviour
- Establish relationships between events



Stage 4: Report the Findings

Forensic reporting is the formal documentation of the investigation methodology, evidence examined, observations, analysis and conclusions.

A forensic report should allow another person to understand:

What was investigated?

What evidence was examined?

How was it examined?

What was discovered?

What conclusion is supported by the evidence?

Important principle

A forensic report should clearly separate:

Fact → Interpretation → Conclusion

An investigator should not present assumptions as established facts.



Chain of Custody

Chain of custody is the documented history of an evidence item from its initial collection through its storage, transfer, examination and final disposition.

It records **who handled the evidence, when it was handled, where it was handled and why it was transferred.**

NIST defines chain of custody as a process that tracks evidence through collection, safeguarding and analysis by documenting handlers, dates/times and reasons for transfers.

Simple example

Evidence collected → Evidence sealed → Evidence transferred → Evidence examined → Evidence stored

Every important transition should be documented.



CHAIN OF CUSTODY



Why Chain of Custody is Important

Digital evidence is particularly sensitive because digital information can be changed or copied easily. Therefore, an investigator must be able to demonstrate that the evidence examined is the same evidence that was originally collected and that its handling was controlled.

Chain of custody provides:

Accountability: Shows who handled the evidence.

Traceability: Shows the movement of evidence.

Integrity support: Helps demonstrate controlled evidence handling.

Legal defensibility: Provides documentation that can be examined if the evidence is challenged.

NIST emphasises that evidence must be properly preserved and that its chain of custody should be tracked throughout its lifecycle.



Chain of Custody: Essential Fields

A chain-of-custody record normally identifies the evidence and documents its movement.

Important information

Evidence Identification What exactly was collected?

Case Number Which investigation does it belong to?

Collector Who collected the evidence?

Date and Time When was it collected?

Location Where was it obtained?

Condition / Seal What was the physical or packaging condition?

Transfer Information Who transferred and received it?

Purpose of Transfer Why was the evidence moved?



What Can Break the Chain of Custody?

The chain of custody can become questionable when evidence handling is not properly documented.

Examples

Missing documentation: A transfer occurred but was never recorded.

Unidentified handler: It is unclear who had possession of the evidence.

Unexplained time gap: There is a period where the evidence location or custodian is unknown.

Unauthorised access: Someone accessed evidence without proper authorisation.

Improper handling: Evidence was altered without documenting the action.

Result: The investigator may have difficulty demonstrating that the evidence remained properly controlled.



Legal Framework for Digital Forensics in India



Legal Framework for Digital Forensics in India

Digital forensic investigations operate at the intersection of **technology, investigation and law**.

The legal framework determines issues such as:

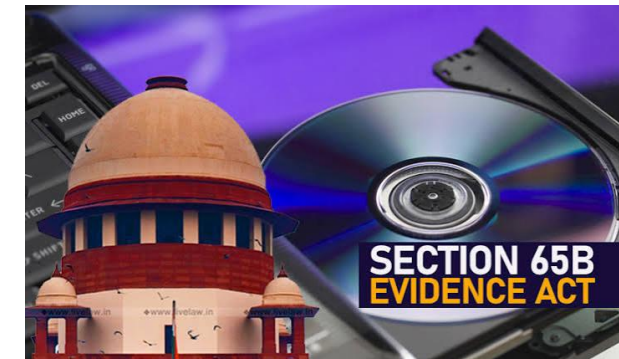
- Authority to obtain information
- Search and seizure
- Electronic records
- Evidence admissibility
- Data protection
- Intermediary responsibilities

Information Technology Act, 2000

Indian Evidence Act - Section 65B

CrPC - Sections 91 and 93

Digital Personal Data Protection Act, 2023



Information Technology Act, 2000

Section 65 - Tampering with Computer Source Documents

This provision addresses tampering with, concealing or destroying computer source documents in the circumstances specified by the Act.

Section 66 - Computer-Related Offences

This section addresses certain computer-related acts when committed dishonestly or fraudulently.

Section 79 - Intermediary Liability

This provision establishes the legal framework governing intermediary liability and the conditions associated with intermediary protection.

Relevance to forensics

Digital investigators may encounter activities involving unauthorised access, manipulation of computer resources, or digital evidence connected with offences covered by the IT Act.

Official source: [Information Technology Act, 2000 - India Code](#)



Electronic Evidence and Section 65B

Section 65B of the former Indian Evidence Act became widely associated with the **admissibility of electronic records** and the certification requirements for computer-generated electronic evidence.

Forensic investigators therefore needed to pay attention to how electronic records were obtained, generated, preserved and presented.

Forensic relevance

When electronic evidence is presented, questions may arise regarding:

- Source of the evidence
- Method of extraction
- Reliability of the system
- Integrity of the record
- Certification requirements

Current legal context

Sakshya Adhiniyam, 2023, which contains updated provisions for electronic and digital records.

Official source: [Bharatiya Sakshya Adhiniyam, 2023 - India Code](#)



Search and Seizure of Digital Evidence

The syllabus introduces **CrPC Sections 91 and 93** in relation to search and seizure.

The underlying forensic principle is that investigators must have appropriate legal authority when obtaining evidence.

Why is this important?

A forensic investigator should not simply access or seize any computer, phone or account because it might contain useful information.

There must be an appropriate basis for:

- Obtaining the evidence
- Searching the relevant location/system
- Requesting information
- Preserving and handling the evidence

Current framework

The **Bharatiya Nagarik Suraksha Sanhita, 2023** has replaced the CrPC framework and includes provisions specifically addressing electronic communications and digital evidence.



Digital Personal Data Protection Act, 2023

The **Digital Personal Data Protection Act, 2023** provides a legal framework for processing digital personal data while establishing obligations and rights relating to such data.

Digital forensic investigations can involve personal information such as:

- Emails
- Messages
- Names
- User accounts
- Documents
- Photographs
- Financial information
- Device-related information

Forensic relevance: Investigators must ensure that personal data is handled only for legitimate and authorised purposes and is protected from unnecessary disclosure or misuse.

Key idea: Investigative access to data does not remove the responsibility to protect that data.

Official source: [Digital Personal Data Protection Act, 2023 - India Code](#)



Role of a Digital Forensic Investigator



Role of a Digital Forensic Investigator

A **digital forensic investigator** is a professional who identifies, preserves, examines and analyses digital evidence using documented and technically defensible methods.

The investigator's responsibility is not to prove that a particular person is guilty or innocent.

The responsibility is to determine **what the available evidence demonstrates**.

Core responsibilities

The investigator:

- Defines the investigation scope
- Identifies relevant evidence
- Preserves evidence
- Performs or supervises acquisition
- Examines forensic artefacts
- Correlates evidence
- Reconstructs events
- Documents findings
- Prepares the forensic report



Qualities of a Good Forensic Investigator

A forensic investigator needs both technical and investigative skills.

Technical understanding :The investigator should understand operating systems, file systems, networks, logs, applications and digital-storage technologies.

Analytical thinking : The investigator must be able to connect individual artefacts and reconstruct a sequence of events.

Attention to detail: Small details such as timestamps, file paths and user identities can significantly affect an investigation.

Objectivity: The investigator must avoid allowing assumptions or personal opinions to influence interpretation.

Documentation: Every important action and observation should be recorded.

Professional integrity: The investigator should report what the evidence supports - **not what someone expects the investigation to prove.**



First Responder

A **first responder** is the person who initially encounters or responds to a system, device or environment containing potentially relevant digital evidence.

The first responder may be a:

- SOC analyst
- IT administrator
- Incident responder
- Security professional
- Police/law-enforcement officer
- Forensic investigator

Primary responsibility

The first responder should **protect the evidence and document the situation before unnecessary interaction changes the state of the system.**



First Responder: What Should Be Documented?

Before interacting with a potentially compromised system, the responder should record the available state as accurately as possible.

Important observations

System state: Is the system powered on or off?

Screen state: What is currently visible?

Connected devices: Are USB drives, phones or other devices attached?

Network state: Is the device connected to a network?

People present: Who is present at the location?

Time: What is the current date and time?

Principle

Document first, interact second.

This reduces the possibility of unintentionally changing important evidence.



Volatile data

Volatile Data is information that can disappear or change rapidly when system power or system state changes. The most important example is **RAM**.

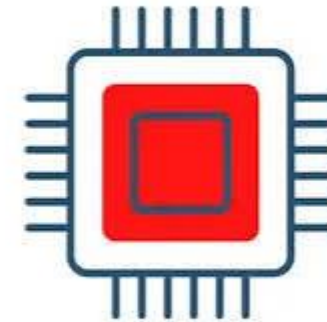
RAM can contain information about:

- Running processes
- Active network connections
- Logged-on users
- Open files
- Commands
- In-memory malware
- Temporary information

Examples of Volatile Memory



RAM Memory



Cache Memory

www.quicklearncomputer.com



Non-Volatile Data

Non-Volatile Data

Non-volatile information remains stored after the system is powered off.

Examples include:

- Hard-drive files
- SSD contents
- Stored logs
- Configuration files
- Archived information

Main difference

Volatile data is time-sensitive.

Non-volatile data generally persists

Examples of Non-Volatile Memory



floppy disk



compact disc



internal hard drive



external hard drive



flash drive



SD card



Order of Volatility

Order of volatility refers to the principle of collecting evidence according to how quickly it is likely to disappear or change. The syllabus provides the following priority:

1. **RAM:** Most immediately vulnerable to loss when power is removed.
2. **Network Connections:** Active connections can change or terminate.
3. **Running Processes:** Processes can terminate or change their state.
4. **Logged-On Users:** Active sessions can disappear.
5. **Open Files:** Application state can change.
6. **Disk:** Stored information generally persists after shutdown.



Why Volatile Data Must Be Considered First

Imagine a computer infected with malware.

While the computer is running, RAM may contain:

Malicious process + command-and-control connection + attacker session

If the computer is suddenly powered off, much of this information may disappear.

Therefore, an investigator must consider the possibility that **shutting down a live system can destroy valuable evidence**.

However, the decision to acquire volatile evidence must consider the investigation circumstances, authorisation, risks and available procedures.

Key principle

The more volatile the evidence, the greater the urgency to consider its preservation.



Integrated Example

Suspected Data Theft

An organisation discovers that an employee may have copied confidential information.
The workstation is still running.

Initial observations

- A confidential file is open.
- A USB device is connected.
- The employee is logged in.
- The machine is connected to the network.

What should the investigator think about?

Locard's Principle

The interaction may have created digital traces.



Integrated Example

Volatility

RAM, active connections and running processes may be lost if the machine is powered off.

Chain of Custody

Every evidence-handling action must be documented.

Investigation Process

The incident must be assessed, evidence acquired, information analysed and findings reported.

Legal Framework

Evidence collection must have appropriate authority and comply with applicable requirements.



Thank you

